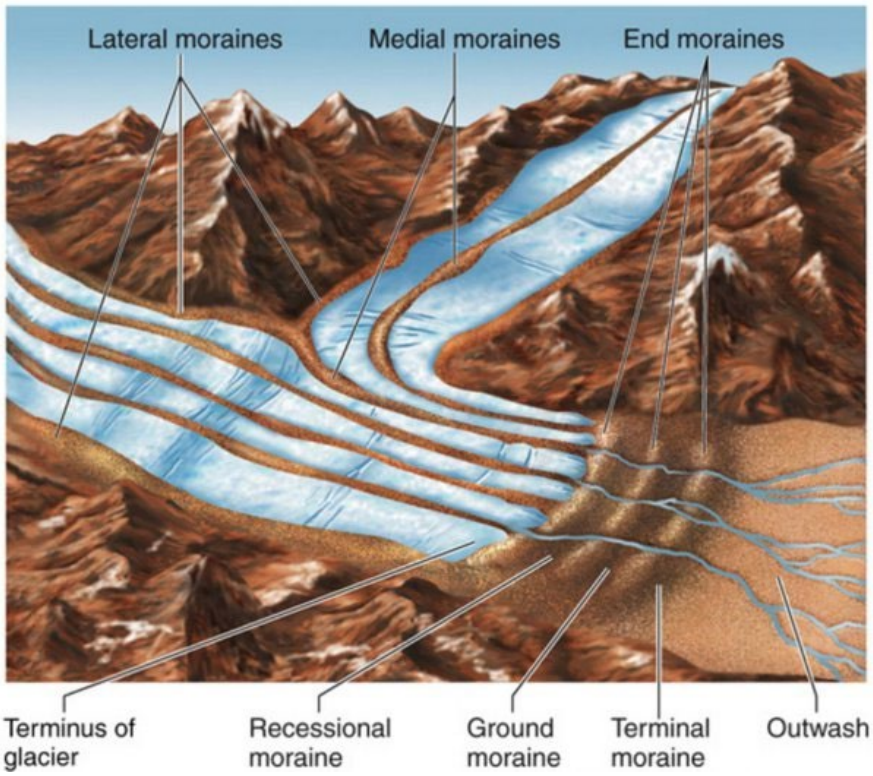


- **The Topic for the next class-** Climatology.

Geography Class 20

Questions on topics taught in the previous class (05:12 PM)

Climatology (05:21 PM)

- **Atmosphere:** The gaseous envelope which **surrounds the earth** from all sides and is attached to the earth's surface by gravitational force.
- **Weather:** It is the day-to-day state of the atmosphere.
- **Climate:** The average atmospheric conditions of an area over a considerable period of time calculated for a **minimum of 30 years**.

Composition of Atmosphere (05:45 PM)

Table 8.1 : Permanent Gases of the Atmosphere

Constituent	*Formula*	*Percentage by Volume*
Nitrogen	N_2	78.08
Oxygen	O_2	20.95
Argon	Ar	0.93
Carbon dioxide	CO_2	0.036
Neon	Ne	0.002
Helium	He	0.0005
Krypto	Kr	0.001
Xenon	Xe	0.00009
Hydrogen	H_2	0.00005

- **Nitrogen:** The atmospheric nitrogen gets converted into soil nitrogen through **nitrogen fixation**.
- Plants are capable of absorbing it from the soil.
- **Oxygen:** Oxygen is **chemically active**. Oxygen is taken directly through respiration.
- It is consumed at the **cellular level**.
- Oxygen is released into the atmosphere through the process of photosynthesis by plants.
- **Carbon dioxide:** It is released by both plants and animals and consumed by plants only i.e. during Photosynthesis.
- It is called **Greenhouse Gas**.
- It helps in maintaining a stable temperature in the atmosphere.
- **Noble Gases:** These gases are **non-reactive** in nature.
- **Water vapour:** It is the most **variable gas** in the atmosphere. It is responsible for all weather phenomena in the atmosphere.
- It is a **greenhouse gas** capable of absorbing Heat.
- 99 per cent of water vapour is concentrated in the lower 16 km of the atmosphere.
- **Dust Particles:** It is concentrated in the lower levels of the atmosphere.
- It acts as **hygroscopic nuclei** and helps in condensation and cloud formation.
- It is released into the atmosphere through weathering, erosion, and from Volcanic Eruptions.

Hygroscopic Nuclei: Hygroscopic means something which attracts water. Hygroscopic Nuclei are microscopic particles that allow water vapours to condense on their surfaces.

Structure of Atmosphere (06:20 PM)

Homosphere	**Heterosphere**
It is a layer with uniform mixing of gases.	In this layer, gases form different layers.
It is up to 80 km.	It is beyond 80 km.
It contains water vapour and dust particles.	It contains the minimum amount of water vapour and dust particles.
In this layer, gases are in molecular form.	In this layer, gases are in atomic form.
It consists of three layers: troposphere; Mesosphere and Stratosphere.	It consists of two layers: Thermosphere and Ionosphere.

Troposphere (06:47 PM)

- It extends from sea level **to 16 kilometres** in the tropics and 6 kilometres in the poles.
- It is a **zone of turbulence** and convection due to the continuous mixing of gases.
- All weather phenomenon is restricted to Troposphere due to the presence of water vapour and dust particles.
- Temperature decreases from sea level **at 6.5 degrees Celsius per kilometre, called** the Normal Lapse Rate.
- The edge of the troposphere is called the **tropopause**, which is a transition zone to the stratosphere.

Stratosphere (07:14 PM)

- It extends from Tropopause to **50 Kilometres.**
- It is called so because the layers are **stratified** and **non-convective** in nature.
- The temperature rises in the stratosphere due to the **absorption of UV rays by Ozone** in this layer.
- The conditions are **extremely dry** except for some rare clouds called stratospheric clouds. Example: Mother of Pearl.
- The top of the stratosphere is called **Stratopause**.

Mesosphere

- It extends from stratopause to **80 km.**
- **Temperature decreases** with height and reaches its lowest in the atmosphere.
- The top of the mesosphere is called Mesopause.

Thermosphere (07:43 PM)

- It extends from **80 km to 500-700 km.**
- It consists of a lower Nitrogen layer and upper oxygen layer.
- The temperature rises rapidly due to the absorption of high-energy solar radiation.
- Temperature reaching up to **1200 degree celsius** at 350 km.

Exosphere

- It extends from the edge of **the thermosphere to the vacuum.**
- It contains atomic Oxygen, Helium, and Hydrogen.
- It contains a **magnetosphere** composed of electrons and Protons derived from Solar Winds and charged particles.
- They have been arranged in two bands at 3000 km and 16000 km which are called as **Van-Allen Radiation Belts.**

Ionosphere

- It exists between **80 to 650 km** within the thermosphere.
- It is formed due to **ionization by solar radiation**. It is made up of multiple layers D, E, F, G, etc
- Ionosphere reflects **short-wave radio waves** helping in radio communication.

Karman Line

- It is the boundary where outer space begins. It is at **an altitude of 100 km** from sea level.
- International law does not define the limit of air space. It is a line agreed upon for aircraft navigation through consensus.
- Above, this line of **air is too thin** for any aircraft balloon or any other flight, other than spacecraft.

(Next Class: Insolation, Heat Budget)

Geography Class 21

INTRODUCTION (5:12 PM)

- Overview & doubts related to the previous class.

INSOLATION, HEAT BUDGET, AND ALBEDO (5:22 PM)

- The amount of incoming solar radiation is called **insolation.**
- The amount of insolation received by the Earth is 2 billionth of solar energy.
- The amount of insolation received on the surface= 1.92 calories per cm2 per minute called the **Solar Constant.**
- The incoming solar radiation is shorter in wavelength.
- This radiation is absorbed by the Earth's surface and is reemitted as **long-wavelength Terrestrial Radiation.**

GREEN HOUSE EFFECT (5:48 PM)

- The atmosphere is transparent to incoming solar radiation and opaque to outgoing terrestrial radiation.
- The warming of the Earth's atmosphere and its surface by the absorption of terrestrial radiation by some of the gases is called the **Green House effect.**
- **The gases responsible are called Green House Gases**, for example, CO2, CH4, N2O, HFCs (Hydro-fluro carbon), PFCs (Per-fluro carbon), SF6 (Sulphur hexafluoride), Water vapor, etc.
- The Earth's surface absorbs incoming solar radiation and emits terrestrial radiation hence it acts as a source of heat for the atmosphere.
- Therefore, temperature decreases with height at the rate of 6.5 degrees celsius per km which is called **the Normal Lapse Rate.**

HEAT TRANSFER (5:56 PM)

- **Methods of heat transfer:**
- **(1) Radiation**
- It involves the transfer of heat in the form of radiant energy.
- The incoming solar energy is in the form of electromagnetic radiation.
- **(2) Conduction**
- It involves the transfer of heat through molecular activity at the zone of contact.
- It occurs at the zone of contact between the troposphere and the Earth's surface.
- **(3) Convection**
- Transfer of heat by vertical movement of the mass of air.
- For Example, the convection of air along the equator.
- **(4) Advection**
- Transfer of heat by horizontal movement of mass.
- For Example, planetary winds and ocean currents.

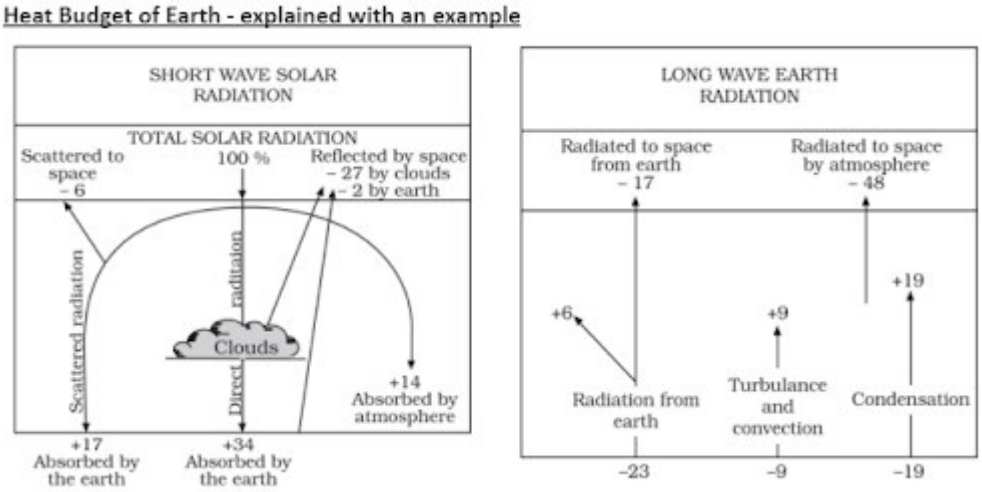
FACTORS AFFECTING INSOLATION (6:16 PM)

- **(1) Transparency of the atmosphere**
- Cloud cover, dust particles, water vapor, etc. reduces the transparency of the atmosphere and reduces insolation received at the surface.
- **(2) Latitude**
- Latitude of a place decides the angle of incidence of insolation which affects insolation received per unit area of the surface.
- **For Example,** in tropical regions insolation falls vertically and is concentrated in a small area. Along temperate and polar regions angle of incidence is oblique and insolation is distributed over a larger area.
- **(3) Length of the day**

- Longer the day, the more the insolation received at the surface. The length of the day depends on the season.

HEAT BUDGET (6:29 PM)

- On the global scale the Earth must reradiate as much heat back to space as it receives from the sun.
- This is necessary in order to maintain a uniform temperature on the Earth.
- The gains and losses in heat by way of incoming solar radiation and outgoing terrestrial radiation are called **Heat Budget.**



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- **Doubts related to heat budget (7:15 PM)**

ALBEDO (7:24 PM)

- Albeo is the ratio between the reflected amount of solar radiation and the incoming solar radiation by the Earth as a whole.
- It is also called as **Reflection Coefficient.**
- The average albedo of the Earth = 35%.

Sample Albedos of Earth

Surface	Typical albedo
Fresh asphalt	0.04
Open ocean	0.06
Worn asphalt	0.12
Conifer forest (Summer)	0.08, 0.09 to 0.15
Deciduous trees	0.15 to 0.18
Bare soil	0.17
Green grass	0.25
Desert sand	0.4
New concrete	0.55
Ocean ice	0.5–0.7
Fresh snow	0.80–0.90

-

TEMPERATURE (7:42 PM)

- The degree of hotness or coldness of a surface is called temperature.

Factors affecting Temperature:

- **(1) Insolation**
- Insolation in turn depends upon the transparency of the atmosphere, latitude, and length of the day.
- Even though the sun's rays are vertical along the equator throughout the year, the equator will not have maximum temperature due to cloud cover.
- **(2) Albedo**
- Surface with higher albedo reflects back more insolation resulting in less absorption of heat causing lower temperature.
- **(3) Nature of the surface**
- Earth behaves differently depending upon the specific heat of the surface.
- The land surface with lower specific heat, heats up more rapidly and intensely than the water surface.
- Also, the land cools rapidly.
- **(4) Distance from the sea or continentality**
- The locations that are in the interior of the continent experience a higher range of temperatures than coastal locations.
- For Example, Delhi experiences hotter summers and colder winters than Mumbai.
- **(5) Distribution of continents**
- Northern Hemisphere with more proportion of land than oceans experience a higher range of temperature than the Southern Hemisphere with more water surface.
- **(6) Altitude**

- Temperature decreases with an increase in altitude.
- **(7) Winds**
- Air in motion is called Winds.
- They transport temperatures prevailing in one area to another.
- For Example, planetary winds.
- **(8) Ocean currents**
- Ocean currents are like rivers inside the oceans.
- They transport warm water from the tropics to polar regions and vice versa helping in the global distribution of temperatures.

TOPIC FOR THE NEXT CLASS: DISTRIBUTION OF TEMPERATURE ACROSS THE GLOBE

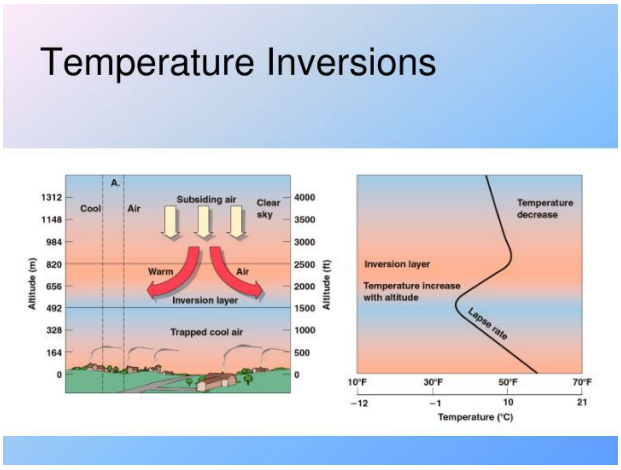
Geography Class 22

Revision of the previous class (05:10 PM)

Temperature and Factors affecting the temperature (05:18 PM)

- **Terms related to the temperature**
- Temperature is the degree of Hotness or coldness.
- The thermometer is used to measure the temperature
- **Diurnal range of temperature**: Difference between daily max and daily minimum.
- **Mean Daily Temperature:** Sum of 24 hours of temperature/24
- Mean monthly temperature: It is the sum of a daily temperature of a month divided by the number of days in a Month
- **Annual Range of temperature**: Highest-Lowest Mean monthly temperature.
- The highest recorded temperature is 58 degrees in Al Azizia (Libya)
- But as per WMO, it is 56.8 in Death Valley (USA California)
- Lowest temperature: Vostok station, Antarctic (-88 degrees)
- **Distribution of Temperature**
- Variation in temperature, Is not uniform along the latitude at the different places due to a number of factors, and when we connect these lines of uniform temperature along the latitude is known as **Isotherm.**
- Across the earth, Isotherms shift north and south with the apparent movement of the sun.
- The Isotherms move northward between January to July and they shift south between July to January.
- The shifting of Isotherms is more in the Northern hemisphere than in the southern hemisphere.
- This indicates higher seasonal variations in the northern hemisphere.
- This is due to the **presence of a vast landmass** in the Northern Hemisphere.
- Moving from land to oceans, Isotherms bents along a particular latitude.
- Isotherms bents polewards in January over the oceans.
- Isotherm bent Equatorward in July over the oceans.
- Near the Equator Annual range of temperature is very less, The Range of temperature increases from the Equator up to higher latitudes, This Range is more in the Northern hemisphere.

Temperature Inversion (05:57 PM)



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- The average rate of decrease in temperature up in the altitude is called as Normal lapse rate
- Under certain special conditions, Reversal of the Normal lapse rate occurs so that temperature increases with altitude. This phenomenon is called temperature inversion.
- The Zone of temperature change in the atmosphere is called a temperature inversion zone.
- **Types of temperature inversion**
- **a)Radiation inversion**
- **The conditions that favour radiation inversion are:**
- Long winter nights
- Calm and windless conditions
- Clear and cloudless sky.
- In these conditions, the land surface radiates heat more rapidly during the night, This loss of Radiation by the land surface causes the land to cool down and the air just above the surface also cools.
- This colder air is overlaid by warmer upper air which has absorbed heat from radiation loss.
- Therefore temperature increases with increases in altitude creating a temperature inversion
- **b) Air drainage type of inversion(06:35 PM)**
- The mountains and hilltops experience rapid fall in temperature during the night.
- The valley sides remain warmer due to radiation exchange between the valleys.
- The cooler and denser air from the hilltops sinks to the valley floor due to Gravity.
- After some time the warmer air rests over the cooler air resulting in a Temperature inversion.
- **c) Advection inversion**
- It is produced when a thick layer of warm air passes over the cold water's surface or snow-covered surface creating temperature inversion.
- **d) Frontal inversion**
- It is caused due to frontal convergence of air masses.
- Along the Fronts, the warmer air is forced upward from the ground by the undercutting of cold air which leads to temperature inversion where warm air is lying above the cold air.

Significance of temperature inversion (06:52 PM)

- Temperature inversion prevents convection by creating stability in the lower atmosphere and hence prevents precipitation.
- The temperature inversion conditions with cold air near the surface are suitable for fog formation.
- It also reduces wind activity and prevents the distribution of pollutants causing bad air quality, For Example: Around Delhi, during winters the pollutants are trapped for a longer duration creating bad air quality.
- In urban areas during winter fog mixes up with smoke resulting in the formation of Smog which is a thick layer and reduces air quality.
- The formation of Fog reduces visibility and disrupts the Transportation network.
- Air Drainage type of inversion decides the pattern of settlements in Mountainous regions, Mountain slopes are preferred over valley floors or hilltops.
- Agriculture in hilly areas is impacted by Air drainage inversion.
- **Example**: Fruit orchards of Uttrakhand and Himachal, Coffee plantation of Brazilavoid valley floors

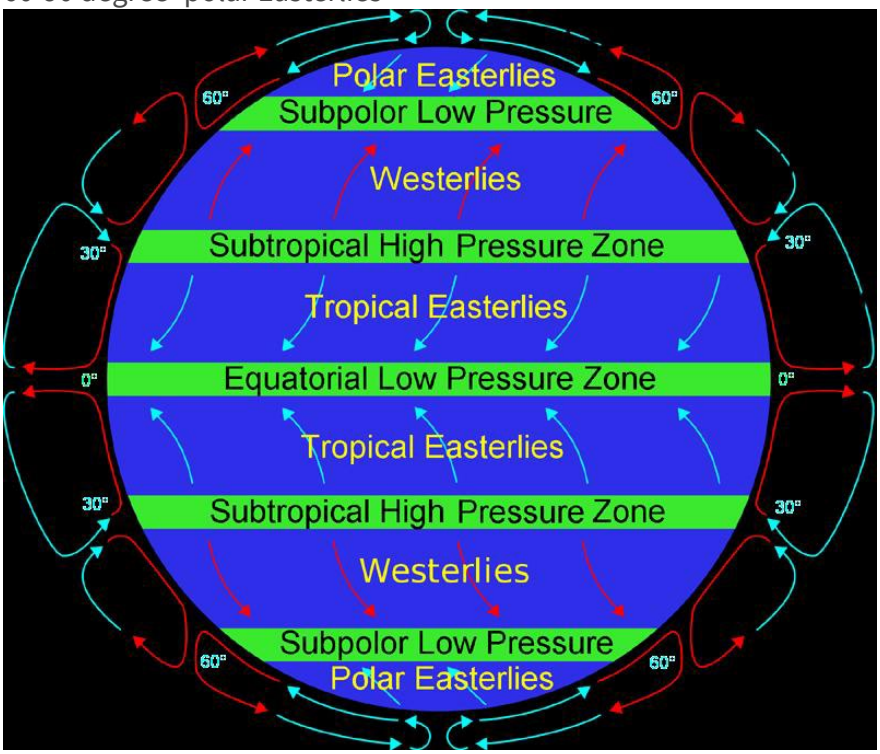
Pressure (07:26 PM)

- Pressure is the force experienced per unit area.
- Atmospheric pressure is the pressure exerted by the Earth's atmosphere.
- The average atmospheric pressure near sea level is equal to 1013milibars.
- Pressure is measured using barometers.
- **Factors that cause variation in pressure**
- **a)Temperature:** When Air is heated it expands and the outward pressure of molecules is spread over a larger area therefore the pressure of air decreases.
- When Air is cooled it contracts to cause an increase in pressure
- **b)Altitude:** The pressure experienced by Air at ground level is higher than that of Air at higher altitudes.
- When Air Rises due to convection its volume increases and pressure decreases.
- When air sinks due to subsidence its volume decreases and pressure increases.
- **c)Rotation:** The rotation of Earth causes air at the poles to be thrown away towards the Equator.
- If we consider the effect of only rotation air piling up around the Equator produces a belt of high pressure and along poles a belt of low pressure.

Formation of pressure Belts (07:53 PM)

- The Belt of low pressure experienced along the Equator region is known as the Equatorial low
- And the Zone of High pressure at the poles is known as Polar High.
- Air Rises at the Equator and moves upward till tropopause and from there separates in two directions i.e two poles.
- At 30 degrees (North and South) it sinks down due to the air being dense now as it is now occupying a smaller area so the pressure increases, so the region is Sub Tropical High.
- From 30 degrees Air then divides into 2 directions again, One moving towards the equator and another moving towards the poles.
- The air near the polar region is always subsiding in nature, Because there is no convection, There is a High-pressure belt so air never vertically rises,
- Air near the polar region gets thrown due to the rotation of the earth and starts to move away from the poles, which earlier was occupying a smaller area now due to

getting thrown away occupies a larger area and at a 60-degree belt has the low-pressure belt. (Temperate low)

- And this air coming from the poles meet the air from the Sub tropics, This convergence causes convection and rises vertically upward and upon reaching the troposphere again divides into two one moving pole side and another towards the subtropics.
- And again meeting the subtropics air at the upper atmospheric level descends at 30 degrees.
- This all results in the formation of Atmospheric cells.
- Between o to 30 degree=Headley cell
- Between 30 to 60 degree= Ferrel cell.
- Between 60 to 90 degree=Polar cell
- Planetary winds flow along the surface.
- 0 to 30 degree =Trade winds
- 30-60 degree= westerlies
- 60-90 degree=polar Easterlies
- 

The Topic for the next class: Winds-Planetary, Local and Seasonal winds.

Geography Class 23

<u>Last class revision(5.10 PM).</u>

<u>Development of pressure belts(5.19 PM):</u>

- Air rising at the equator due to high temperature causes expansion and therefore the development of a low-pressure belt at the equator.
- This connecting air at the equator spreads out as it reaches the top of the troposphere and moves toward the poles.
- This air as it converges towards the polar region its density increases and begins to sink which is subsiding, leading to a high-pressure belt at 30 degrees N and S.
- Some of the high-pressure air moves toward the equator and some toward the pole.

- The air moving towards the equator replaces the air rising there completing a cell.
- Low temperatures at the poles, resulting in a contraction of air and the development of high pressure.
- Air blowing away from poles spread out to larger space and pressure falls leading to a low-pressure belt along 60% north and south.
- Some of the air from the sub-tropical high-pressure belts moving toward the pole reaches 60 degrees north and south and converges with air from the poles leading to convection along 60 degrees north and south.
- The rising air 60 degrees north and south spread out at the edge of the troposphere, a part of it moves towards the equator and the rest towards the poles.

Distribution of pressure across the world(5.41 PM):

- The distribution of pressure across the earth can be studied using isobars.
- Isobars are the lines joining places with equal pressure.
- With the apparent movement of the sun between the tropics, the pressure belts shift north and south.
- ***Pressure in January:***
- The equatorial low-pressure belt extends into the southern hemisphere.
- The sub-tropical high-pressure belt of the southern hemisphere is found only over the oceans.
- The low temperature in the northern hemisphere produces a continuous high-pressure system, linking it with the high-pressure cells of Siberia and North America.
- ***Pressure in July:***
- Equatorial low-pressure belts extend too much into the northern hemisphere due to intense heating in India and Tibet.
- Sub-tropical high-pressure belt in the northern hemisphere is not continuous and exists only over oceans as weak high-pressure cells.
- In the southern hemisphere, a continuous belt of high pressure is formed.

Winds(6.10 PM):

- The wind is air in motion.
- *Forces affecting the motion of wind:*
- ***1. Pressure grading force:***
- The rate of change of pressure concerning distance is called a pressure gradient.
- The pressure differential in the atmosphere causes the movement of air from high pressure to low pressure.
- The force causing this movement is called the pressure gradient force.
- Pressure gradient force acts in a direction perpendicular to isobars.
- Pressure gradient forces increase with the increase in the rate of change of pressure.
- That is pressure gradient force is more when isobars are placed closely.
- ***2. Frictional force:***
- Any moving object moving near land experiences a frictional force in the direction opposite to that of its movement.
- It is maximum near the surface than at higher altitudes.
- It is maximum over the continents than oceans.
- ***3. Coriolis force:***
- It is a fictional force produced due to the rotation of the earth

- It is the combined effects of various forces and factors such as centrifugal force, angular velocity, and variation in speed of rotation at different latitudes.
- It causes any moving object to defect towards the right in the northern hemisphere and the left in the southern hemisphere.
- The degree of Coriolis force depends on the speed of the moving body, the latitude at which the body is moving, speed of rotation of the earth.

Geostrophic winds(7.17 PM):

- At some latitude where the isobar is straight and there is no friction, the pressure gradient force is balanced by Coriolis force and the result wind flows n a parallel direction of isobars. This is called geostrophic winds.
- Geostrophic winds along low-pressure and high-pressure cells result in the formation of cyclonic and anticyclonic circulations.
-

Pressure system	Pressure condition at the center	The pattern of winds in the north Hampshire	The pattern of winds in the South Hampshire
Cyclone	Low	Anticlockwise	clockwise
Anti-cyclone	High	clockwise	Anticlockwise

Planetary winds(7.37 PM):

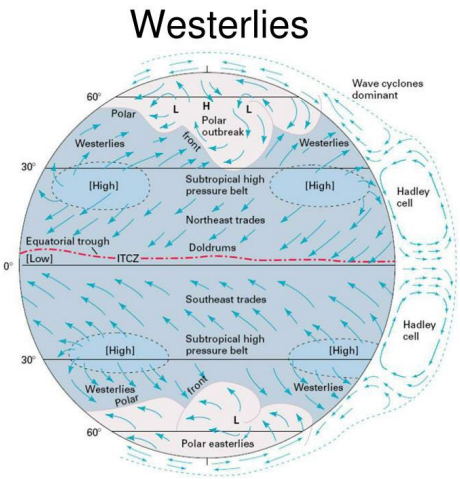
- These are the winds blowing throughout the year and are found across the entire planet.
- ***Trade winds/Easterlies:***
- They blow from subtropical high to equatorial low.
- In German trade means Track as they blow in the same direction, throughout the year.
- Trade winds are dry and stable in the area of origin.
- Moving toward the equator picks up moisture and causes precipitation along the eastern margins.
- Reaching the western margins, as they are devoid of moisture the offshore trade winds will not result in precipitation causing desert formation.
- In addition to this nearness to the sub-tropical high-pressure belt and pressure belt and presence of cold ocean current creates dry conditions, eg Sahara desert, Kalahari deserts, etc.

The topic of the next class: Planetary winds to continue.

Geography Class 24

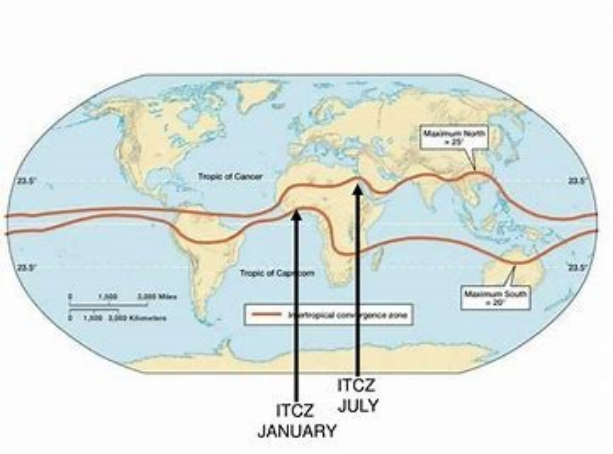
Revision of the previous class (05:07 PM)

Westerlies (05:21 PM)



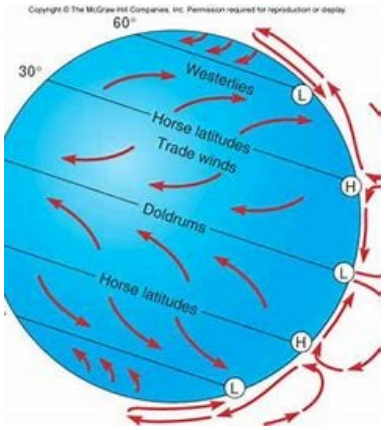
- They blow from Subtropical high to temperate low in both hemispheres.
- The vast landmass in Northern Hemisphere obstructs the Westerlies whereas, in the Southern hemisphere, they blow strongly and consistently due to the huge water expanse.
- Therefore they are called as Roaring forties, Furious fifties, and shrieking sixties in the southern hemisphere.
- **Polar Easterlies**
- It blows from a polar high-pressure belt to a temperate low-pressure belt.
- They are extremely cold, stable, and dry.
- And they blow for long distances and affect the climate.

Zones of Planetary winds (05:30 PM)



- **Inter-Tropical Convergence Zone(ITCZ)**
- Convergence of Trade winds from both the sides of tropics.
- It is centred on the Equator.
- 5 degree North and south of the Equator normally during Equinox, But shift as well North and southwards.
- This zone experiences low-pressure, calm, and windless conditions therefore it is called as the **Zone of Doldrums.**
- Summers: ITCZ Shift toward North of the Equator.
- Winter: ITCZ shift toward the south of the Equator.
- Humidity will be greater due to Convection.

Horse Latitude (05:41 PM)



- 30 Degree North and South.
- It is a Zone of Sub tropical high-pressure belt near 30 degrees North and south.
- Due to vertically descending belts, This is the region of light and calm winds.

Tri-cellular Meridional circulation

- The Surface winds blow from High to Low-pressure belts but in the upper atmosphere, the direction of air circulation is opposite to surface winds.
- This together with convection and subsidence along low-pressure and High-pressure belts result in three cellular circulations along each meridian called Tri cellular meridional circulation.
- The three cells are the Headley cell, the Ferrel cell, and the Polar cell.

Seasonal winds (06:00 PM)

- They are not flowing Throughout the year
- And do not cover the entire planet
- The seasonal differences in Temperature and pressure cause the movement of Air and subsequent winds blowing in particular seasons are called Seasonal winds.
- Example: South West Monsoon.
- **Local winds**
- They are a regular phenomenon
- Differences in the heating and cooling of the Earth's surfaces and the cycles that develop daily or annually create several common winds called local winds.
- **Land and sea breeze**

- The land gets heated up more quickly than the adjacent sea during day time.

- Low pressure is developed over the Land and High pressure over the sea.
- This causes the circulation of relatively cool air from the sea to adjacent land called sea breeze.
- Rapid loss of Heat from the land causes a **reversal** of daytime pressure conditions due to High pressure on land and low pressure over oceans.
- Winds blowing from land to sea cause Land breezes.
- Fishermen along coastal regions use land and sea breeze for fishing activity.
- **Mountain and valley breeze**
- During Day time mountains' slope heats up rapidly compared to the valley.
- This results in the air from the valley moving up along the slopes called as valley breeze or **Anabatic winds.**
- At night the temperature difference between Mountain slopes and valleys is reversed causing winds to blow from mountain to valley causing Mountain breeze which is also called **Katabatic winds.**
- **Tips and tricks to outline the world physical Map (06:47 PM)**

Local winds of the world in Map (07:28 PM)

- **Hot winds:**
- **Chinook:** Western **North America** along the Rockies mountain from the mountain towards the plains in a west-to-east direction
- Significance: Also called a snow eater when the entire Rockies area, Prairies land is covered with snow where wheat is grown in winter so snow melts which is helpful for wheat cultivation during winter.
- **Fohn:** Along the Alps in **Europe** which flow south to North direction.
- Significance: It helps in the melting of snow in Germany, similar to chinook.
- **Harmattan:** The region is **Nigeria** and flows from Northeast to southwest, from land to sea.
- Significance: Harmattan is dry winds that cut moisture when flowing from the Humid region of Africa, This is the reason it is also called **Doctor winds.**
- **Haboob:** Hot winds of **Sudan** which have no specific direction
- **Sirrocco:** Winds from Sahara to the Mediterranean region and then crossed the Mediterranean sea reaches Europe, hits the mountain, and causes orographic Rain. Also called Blood Rain as the winds from the Sahara come along with the Red Dust of the Sahara.
- **Khamsin**: Sirocco is known as Khamsin in Egypt, and Gibli in Libya, All the characteristics same as Sirocco.
- **Berg:** In South Africa, East to West From Land to the ocean.
- **Samoon:** In Iran, and Iraq, No specific direction, Also called poison winds.
- **Loo**: North India, Direction west to East.
- **Karaburan:** Mongolia and Northern China, Direction: Northeast to the southwest.
- **Brickfielder:** In Southern Australia, Direction: North to south
- **Cold winds (07:49 PM)**
- **Blizzard:** North America, particularly in USA and Canada.Direction from North to south.
- Blizzard in Siberia is called Burran.
- **Norte/Northe:** Direction North to South, Region USA, and Mexico respectively.
- **Pamperos:** In Argentina, South to North
- **Mistral**: Region France, Direction North to south.
- **Levant:** Southern Spain, Direction East to west.

- **Bora:** Along the coast of the Adriatic sea from North East to Southwest direction.
-

The topic for the next class: Humidity, Continuation of Climatology.
Geography Class 25

Revision of the previous class (05:12 PM)
Humidity, Evaporation, and Condensation (05:20 PM)

- The amount of water Vapour in the atmosphere is called Humidity, It helps in energy balance on the surface of Earth through Evaporation and Condensation.
- Humidity is the index of the atmosphere's potential for Yielding precipitation.
- **Measurement of Humidity in the atmosphere**
- **3 ways:**
- **a) Specific humidity**
- It is the ratio of the weight of water vapor in grams to the weight of air in kgs expressed as grams/kg of Air.
- It indicates the actual amount of Moisture present in the air.
- Here amount of water vapor remains the same, It is not showing the effect of temperature.
- **b)Absolute humidity**
- It is the ratio of the weight of water vapor in grams to the volume of air in a meter cube
- Expressed as grams per meter cube of air
- Measures humidity in grams/m3 i.e 100 gms/m3
- It decreases with the expansion of air and increases with the contraction
- **c)Relative humidity**
- Measures Humidity in percentage
- It is the ratio between the amount of water vapor present in the air to the maximum amount of water vapor it can hold at a given temperature.
- Relative humidity seen just with temperature.
- If the temperature is increased the maximum capacity increases and relative humidity decreases.
- If the temperature is decreased maximum capacity decreases Relative humidity increases.
- Air is said to be saturated when Relative humidity reaches 100%.

Evaporation (05:58 PM)

- The conversion of liquid water into water vapor is called Evaporation
- **3 important factors**
- a) Higher the temperature, the Higher the rate of evaporation.
- b) Humidity: Higher the humidity, the Lower the rate of evaporation.Due to the saturation of Air.
- c) Wind speed: Higher the wind speed, the Higher the rate of evaporation as Air movement provides a constant supply of fresh air.
- **Distribution**
- Highest in 30 degrees
- Lowest in the polar region.

Condensation (06:07 PM)

- The process of conversion of water vapor into liquid water is called as condensation.
- It is the reversal of evaporation.
- The temperature at which condensation takes place is called a dew point.
- If condensation happens below zero degree celsius it is called a frost point.
- **Condensation forms**
- **Dew**
- It is the moisture deposited in the form of liquid water droplets on land surfaces.
- The conditions favorable for the formation of dew are long winter nights, Calm air, and cloudless sky.
-
- **Frost**
- It is a thin layer of ice on a solid surface.
- It is formed when the temperature of the surface is below the freezing point and the water droplets are falling on such a cold surface.
-
- **Rime**
- The deposition of needle-like white opaque icy crystals on the surfaces with temperature below zero degrees Celcius.
- It occurs when supercooled water droplets in the air come in contact with the surface below the freezing point.

-

- **Fog (07:05 PM)**
- It is produced near the surface when the temperature of the air drops suddenly.
- The visibility will be less than 1 km.
- The condition required for the formation of fog is the same as the condition of temperature inversion
- Following conditions favor fog formation commonly
- Long winter night, Calm air, and cloudless sky.
- **Types of Fog:**
- Radiation fog, Valley fog, Frontal fog, Advection fog
- **Mist**
- It consists of small droplets of water suspended in the air.
- It is similar to fog but with a lower density of water vapor.
- Visibility in Mist is between 1-2 km.
-
- **Haze**
- It is caused by smoke and dust particles with humidity.
- It is generally observed in Urban areas.
- Visibility in Haze is up to 2 km.

Clouds and precipitation (07:13 PM)

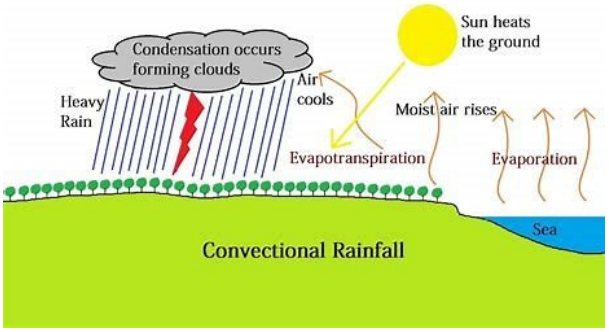
- **Stability condition**
- It is the condition when air resists vertical movement and remains in its original position.
- Stability is when the air is cooled at its base or when air subsides along high-pressure belts.
- Pressure is unlikely in this condition.
- **Instability condition**
- It is a condition where air does not resist vertical movement and leads to cloud formation and precipitation.
- Instability occurs along the regions of High temperature and low pressure.

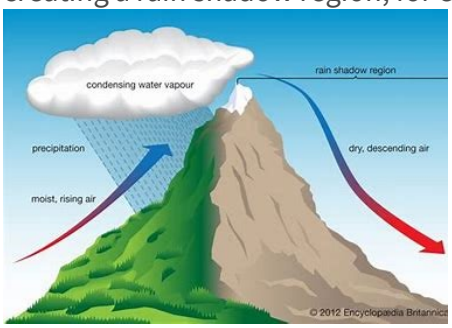
Types of Clouds (07:26 PM)

- **Cirrus**
- Cirrus means Thin, High altitude, and feathery appearance
- White in color and indicates fair weather.
- Ciro stratus
- Layered and High altitude with a milky appearance.
- **Stratus**
- Stratus means layered.
- **Alto**
- Alto means middle altitude.
- **cumulus**
- Cumulus clouds have a cotton wool-like appearance with dome-shaped cauliflower top
- Also called globular masses
- **Nimbus**
- Nimbus means Rain bearing
- **Alto Cumulus**
- Globular masses of clouds with a cotton wool-like appearance at middle altitude.
- **Altostratus**
- A layer of clouds in sheets along the middle altitude.
- **Stratus**
- A low uniform layer of cloud near the ground level produces light drizzle.
-
- **Cumulonimbus clouds**
- Overgrown cumulus clouds are very dark heavy and dense with an anvil top.
- It causes heavy rainfall with thunder and lightning.
-

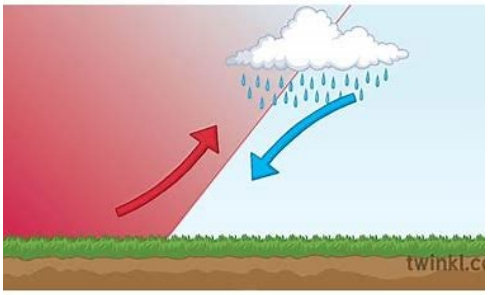
The Topic for the next class: Precipitation, Type of Rainfall.
Geography Class 26

Revision of the previous class (05:13 PM)
Precipitation (05:28 PM)

- Precipitation requires the following **three condition**
- a)Upliftment mechanism: To cause moist air to rise
- b)Saturation and cooling of air below the due point
- c)Presence of Hygroscopic nuclei such as dust particles around which water droplets can accumulate.
- **Types of precipitation**
- **a)Convectional Rainfall**
- It occurs in the region of intense heating near the ground surface.
- Heating causes air to expand and rise resulting in the upliftment.
- They are often accompanied by thunderstorms and lightning due to the formation of cumulonimbus clouds.
- They are experienced throughout the year in equatorial regions and in summers in tropical regions.
- 

- **b)Orographic Rainfall**
- When warm and moist air is forced to rise across mountain slopes it cools down causing precipitation along the windward side.
- However, on the leeward side, the descending air will not cause precipitation creating a rain shadow region, for example, The Western Ghats of India.
- 

- **c)Frontal Rainfall**
- It occurs along frontal zones due to the convergence of different types of air masses.
- The warm air mass rises above the cold air mass causing precipitation.
- Frontal precipitation is common along the temperate region.

- 

- **Distribution of precipitation**
- Along the Equatorial region, the highest precipitation is experienced above 200 cm per annum.
- It is due to the convectional Rainfall
- Along subtropical regions around 30 degrees minimum rainfall experienced 25 cms per annum.
- The Mid-latitude regions between 30 to 40 degrees of North and south experience uniform and moderate precipitation due to the frontal rainfall.
- Due to the presence of moist maritime air mass oceans receive more precipitation than the continents.
- Coastal regions receive higher precipitation than the interior.

Thunderstorm and Lightning (05:49 PM)

- Due to intense heating and strong vertical convection, Cumulonimbus clouds are formed.
- The Raindrops in these clouds move up and down due to strong air currents creating electric charges which accumulate on opposite sides of clouds.
- When both types of charges have attracted a flash of light is produced due to the exchange of charges called lightning.
- Lightening causes Vaccum in the cloud due to the Rapid expansion of air.
- It is filled by the surrounding cold air producing **claps of Thunder.**

Cloud Burst (06:10 PM)

- IMD defines cloud burst as Rainfall over 10cm per hour concentrated in a small area of a few km.
- **Formation of cloud burst**
- High temperature and higher humidity favour strong and rapid convection.
- Steep topography along the mountainous region causes a huge buildup of clouds without precipitation due to the funnelling effect.
- The Rapid convection prevents the Raindrops to fall down.
- Once the cloud becomes too heavy for the air to hold on it drops together in quick succession resulting in a cloud burst.
- In India, cloud bursts can be observed along the Himalayas, Rajasthan desert, and monsoon coasts.

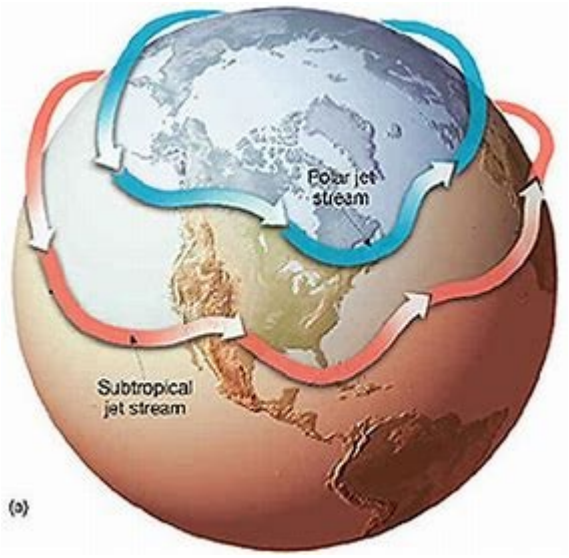
Tornado (06:36 PM)

- A violently rotating column of air extends from the ground to the base of a Thunderstorm.
- It appears like a funnel descending from a cloud.
- The wind speed goes up to 400 km per hour.
- It is formed with changes in wind speed and direction creating a horizontal spinning effect within a storm cell.
- This effect is tipped vertically by rising air moving up through the thunderclouds.
- A water spout is a whirling column of air and water mist developed over water bodies.

Jet streams (07:11 PM)

- **WMO** defines jet streams as a strong narrow current concentrated along a quasi-horizontal axis in the upper troposphere or lower stratosphere characterized by strong vertical and lateral wind shear.
- Featuring one or more velocity maxima.
- **Characteristics:**
- They are thousands of km in length and 100 km in width.
- Normally blows from west to east.
- winds are faster during winter.
- The entire system follows the sun.
- It follows the Zig Zag path in the form of Rosby waves.
- **Causes**
- Jet streams are generated due to the temperature difference between the two regions that is Tropical and Temperate, Temperate and polar, etc.
- The difference in pressure gradient with altitude above cold and warm air mass causes strong movements of winds.
- The larger the temperature difference, the stronger the winds.

Types of Jet streams (07:49 PM)



- a)Polar front jet streams.
- They are formed where the polar and Ferrel cells meet.
- They are irregular and discontinuous.
- They blow from west to East.
- b)Subtropical westerly jetstream
- Associated with temperature gradient between Headley and Ferrel cell.
- It is more regular and strong, It blows from west to east.
- c) Tropical Easterly Jet stream
- Forms over India and Africa only in summer due to the intense heating of the Tibetan plateau.
- The direction is east to west.
- d)Polar Night jet stream
- Formed over the polar region above the troposphere during winters.
- They are associated with the formation of the ozone hole.
- The direction is from west to east.
- e)Local jet stream
- They are formed due to local thermal and dynamic conditions.
- Example: Somali jet streams
- **Significance of the Jet streams**
- The jet stream is responsible for cyclonic and anticyclonic conditions along the surface.
- Upper divergence in jetstreams causes lower convergence and upper convergence causes lower divergence.
- It changes the weather condition along the surface by suppressing or increasing precipitation.
- It is used in Aircraft navigation.
- The polar night jetstreams are responsible for Ozone depletion.
- The jet streams help in the transportation of pollutants into the upper troposphere from urban regions and help in clearing fog and smog during winter.
- It helps in the formation of temperate cyclones and intensifies them.
- The monsoon in south Asia is controlled and affected by subtropical westerly, Tropical easterly, and Somali jet streams.

- The sub-tropical westerly jet streams help in bringing western disturbance causing precipitations during winters in the Northwest region.
- The interactions of jet streams with la Nina conditions cause the formation of a heat dome over North America.
- Meandering of Jetstreams near polar regions during winters results in the outbreak of polar vortex causing freezing conditions along lower latitudes.

The topic for the next class is: Polar vortex, Continuation of climatology

Geography Class 27

Revision of the previous class in a detailed way (05:05 PM)

- **Polar Vortex (05:30 PM)**
- Polar Vortex is a large area of low pressure and cold air surrounding the Earth's pole.
- The term Vortex refers to the counter-clockwise flow of air which keeps the cold polar air locked inside.
- The polar winter night jetstreams keep this region covered with strong winds.
- Occasionally when the vortex weakens due to the weakening of jetstreams the vortex expands and sends the cold air southwards along with jetstreams.
- This brings down the temperature in southern Regions to colder levels of below zero degree Celsius.
- **Air masses (05:47 PM)**
- It is the extensive portion of the atmosphere.
- Whose physical properties such as temperature pressure and Humidity are homogeneous horizontally and vertically for hundreds of km.
- The temperature of the source area shall remain uniform horizontally with a very gradual change in the vertical direction.
- The humidity conditions should also be uniform with the least variability.
- The surface area shall remain homogenous for a few hundred km.
- The Regions with light divergent winds, High-pressure belts, and anticyclonic circulations are more favorable for the formation of air masses due to stable conditions.
- **Types of Air masses**
- Classified based on different factors such as:
- a) Source area:Continental(c) formed over the continents and Maritime(m) formed over oceans
- b) Temperature: Warm and Cold air mass
- c) Pressure: Stable(s) and Unstable air mass(u)
- d) Latitude:Arctic air mass(a) and Antarctic air mass(aa) Polar air mass(P) Tropical air mass(T0 Equatorial air mass(E)
- a and aa is continental
- P & T is continental and maritime respectively.
- E is maritime.

Significance of Air mass (06:16 PM)

- **Air masses** Transport heat from one region to another and help in the global distribution of temperature.
- When Air mass moves away from its source area it changes the temperature and pressure conditions of the destination regions.

- Maritime air masses bring More moisture toward the continental region increasing its humidity and causing precipitation higher precipitation along the coastal regions due to the influence of Maritime air masses.
- The dry Continental air masses along the subtropical regions help in maintaining the aridity conditions.
- The cold air mass formed over Siberia and Canada affects the surrounding region for long-distance causing cold conditions.
- The maritime air mass of the Indian ocean plays an important role in precipitation over India during Monsoon.
- The interaction of warm and cold air masses in temperate regions leads to the formation of fronts and temperate cyclones.
- The maritime tropical and equatorial air masses in the Atlantic and Indian oceans help in the development of Tropical cyclones.
- **Temperate Cyclones (06:56 PM)**
- **Front**
- A narrow zone of transition dividing two air masses of differing temperature and humidity conditions is called the front.
- They are usually formed in Mid latitude or temperate regions.
- The process of formation of fronts is called frontogenesis
- The process of decay of fronts is called as
- **Types of Fronts (07:07 PM)**
- **a) Stationary front**
- It is a front where the surface position of a front is not changing and the two air masses are unable to push each other.
- When a stationary front is disturbed it results in a warm front or cold front.
- **b) Cold front**
- It is the boundary between warm and cold air mass where the cold air mass is advancing and undercutting the warm air mass.
- It results in a steeper gradient along the front leading to rapid upliftment of warm air and the formation of cumulonimbus clouds.
- It results in heavy rainfall over the smaller regions.
- **C) Warm front**
- It is the boundary between warm and cold air mass where the warm air mass is advancing and overriding the cold air mass.
- The gradient of the front is gradual resulting in strato nimbus clouds.
- It causes moderate rainfall over a larger area.
- **Occluded Front (07:39 PM)**
- It is the later stage of front formation where the air in the warm sector is no longer at the ground surface due to the meeting of a warm and cold front.
- It is a compound zone with both warm and cold front characteristics.
- The occluded front gradually results in frontolysis.
- It is the later stage of front formation where the air in the warm sector is no longer at the ground surface due to the meeting of a warm and cold front.
- It is a compound zone with both warm and cold front characteristics.
- The occluded front gradually results in frontolysis.
- **Formation of Temperate Cyclones (07:45 PM)**
- 6 Stages

- Stage 1:Convergence of two air masses with different characteristics. Air masses move parallel to each other and a stationary front is formed.
- Stage 2: Warm and cold air masses penetrate into the territory of each other forming a wave-like the front.
- Stage3: With warm and cold fronts starting to move towards each other, The warm sector is wedged between the cold sector.
- Stage 4: The warm sector is narrowed in extent due to the advance of a cold front.
- Stage 5: The occlusion starts with warm air rising completely above the cold air and the cold front completely overtaking the warm front.
- Stage 6: The warm sector completely disappears, the occluded front is eliminated and the cyclones die out.
- Temperate cyclones are also called Mid-latitude cyclones or extratropical cyclones or wave cyclones.
- They are more prominent in the regions between 35-65 degrees North and south.
- **Role of Jet streams in Temperate Cyclones**
- **Jet streams** Being Quasi horizontal in nature results in upper divergence in some locations and upper divergence causes surface convergence which helps in bringing together the warm and cold air masses and the subsequent formation of fronts.

The topic for the next class: Tropical cyclones

Geography Class 28

Revision of the previous class and Doubts are taken (05:12 PM)

Tropical cyclones (05:40 PM)

- A Tropical cyclone is a **low-pressure high velocity** wind system originating within the tropics over the oceans.
- Names in different regions:

Indian Ocean	Cyclone
Atlantic ocean	Hurricane.
South china sea	Typhoon
Japan	Taifu
Philippines-	Bagui
Australia	Willy Willy

Stages in Tropical cyclone (05:58 PM)

- **7 stages of Tropical Cyclones**
- **a)Temperature:** Above 27-degree celsius over the ocean to ensure sufficient evaporation
- **b)Depth of warm water:** The 27-degree Celcius temperature should extend up to 60-70 meters in depth.
- **c)The humidity of Air:** High Humidity of 50-60 % near the surface.

- **d)Minimum vertical wind shear:** So that Humid and warm air is not swept away.
- **e)Coriolis force:** It causes wind deflection therefore cyclones are developed only between 5 degrees North to 25 degrees North or 5 Degree south to 25 degrees south.
- **f)Pre Existing Low-pressure conditions:** These get intensified to form Cyclones.
- **g)Upper Air divergence:** Causing convergence near the surface.
- In the above, the First 5 are the necessary condition.
- **Formation**
- A strong Convection results in the build-up of huge cumulonimbus clouds
- The Release of Latent heat provides more energy to the system supporting further strong Convection.
- When the wind starts to descend along the sides the entire system is stabilized.
- The Lateral winds near to the surface start to Rotate around the Low-pressure centre due to the Coriolis force.
- After some time some of the dense air rising near the centre starts to descend resulting in the formation of an eye.
- Once a cyclone is formed it moves under the influence of planetary winds and deflects toward the northwest direction.
- When a cyclone makes landfall or enters the waters of higher latitude, It dissipates and comes to an end.

Structure of the Cyclone (07:18 PM)

- **Eye:** It is the centre of the storm about 5-50 km in diameter with a clear sky, Calm conditions, Lowest pressure, and Highest temperature.
- **Eye wall:** A wall of thick cumulonimbus clouds with 10-20 Km of width. It has the strongest wind in the cyclone. Thunderstorm occurs in this region due to intense convection.
- **Spiral bands:** AKA rain bands or feeder bands extend for 100 km and lead to Galaxy like appearance.
- **Annular Zone:** A Zone of suppressed cloudiness, High temperature, and low humidity.
- **Outer convective bands:** At the edge of the main cloud mass with instability and convection

Naming System of Cyclones (07:25 PM)

- In 2004, 8 countries of the Northern Indian Ocean agreed to have a common naming system for Cyclones. Each of the countries arranged them in alphabetical order.
- And it was submitted in a set of 8 names.
- The Regional Specialized Meteorological centre of WMO located in New Delhi maintains the list arranged in sequential order and assigns the names when the wind speed crosses 62km/hr.
- The suggested names shall be neutral politically, Religiously, and Culturally as well as concerning Gender.
- It should be easy to pronounce.
- All the names in the earlier matrix, Expired in 2019, and a new matrix suggested by 13 countries was agreed upon effective from 2020.

Recurvature of the cyclones (07:46 PM)

- When the Air currents in the local atmosphere push the air from the poles towards the Equator and interfere with Cyclone formation the Cyclone which is about to

diminish gets more wind and deflects right or eastward under the influence of westerlies this phenomenon is known as Recurvature of the cyclones.
- Recurvature is very difficult to predict, for example, Cyclone Ockhi of 2018.

Cyclones in the Indian Ocean (07:52 PM)
- The number of Cyclones in the Bay of Bengal is more than in the Arabian sea in a Ratio of 4:1
- **Reasons:**
- Strong Wind shear in the Arabian sea
- Temperature is higher in the Bay of Bengal because more rivers enter the Arabian sea.

Colour-coded warning of IMD for Cyclones (07:57 PM)
- It is used by IMD to signify the intensity of the situation and the warning associated with it
- **Four colours are used:**
- Green-No warning
- Yellow-Be aware
- Orange-Be prepared.
- Red-Take Action
- The objective is to alert people of Hazardous weather and prepare them for handling disaster situations.
- It is Universal and issued during floods and heavier Rainfall for cyclones it is represented through a matrix.

Difference between Tropical and Temperate cyclones (08:01 PM)
- You Can Refer to a chart Drawn in the class

The topic for the next class: Climatic Regions

Geography Class 29

Climatic regions: (5:25:57 PM):
- 0 to 10 degree:-Hot and a wet equatorial type of climate.
- 10 to 30 degree:-Hot zone includes desert, savanna climate, and Monsoon type of climates.
- 30 to 45 degrees- Meditterean, steppe, and China type of climate.
- 45 to 60 degrees-Coll temperate zone Includes British, Siberian, and Laurentian types of climate.
- 66 1/2 degrees and above-Arctic zone.

Hot and wet equatorial climate:(6:04:41 PM):
- Temperature is always high throughout the year.
- Precipitation happens through convection.
- The daily maximum temperature is 30 degrees and the daily minimum is 20 degrees.
- The diurnal range of temperature is more than the annual range and both are very less.
- Humidity levels are high.
- No marked differences between the seasons.
- Precipitation above 250 cm through conventional rainfall.
- Life is rich here.>> Vegetation is evergreen in nature.
- Evergreen rainforests exist here.
- Vegetation shows the layered arrangement of trees.

- Sunlight is the limiting factor.
- The diversity of flora and fauna is very high.
- Regions- Amazon basin, Parts of Colombia, Ecuador, Congo Basin, Indonesia, Malaysia, Singapore, Papua New Guinea.

Monsoon type of Climate:(6:25:10 PM):

- Hot zone eastern Margin.
- Seasonal reversal of winds.
- Precipitation is concentrated during the monsoon period.
- One or more months of the dry season.
- Vegetation is a deciduous type.
- Trees shed leaves during winter.
- Regions- South Asia including India, Pakistan, Bangladesh, Indo-China, Eastern Africa, and Northern Australia.
- Within the monsoon type, onshore trade winds cause precipitation from both directions throughout the year called a tropical marine climate.
- Examples- Central America, SE Brazil, etc.

Sudan type of climate:

- Hot zone continental type of climate.
- Grasslands in tropical regions.
- Precipitation is lesser in the eastern margin and more than in the western margin.
- Rainfall in summer due to convection.
- The definite dry season during winters.
- The average annual rainfall is 70 cm.
- Scattered drought-resistant trees.
- Regions- Sudan region, Savanna region to the south of the equator in Africa, South America-Llanos, and Campos and North Australia.

Desert Type of climate:(7:10:00 PM):

- Hot zone western climate.
- **Formed by 3 conditions;**
- -Nearness to the subtropical high-pressure belt.
- -Offshore trade winds.
- -the presence of cold ocean currents.
- Precipitation is very less with less than 25 cm per annum.
- High Diurnal range of temperature.
- Vegetation is xerophytic in nature.
- Eg- Cactus, dates, etc.
- Regions-Sahara desert, Mohave desert, Arizona desert, Atacama desert, Namib desert, Kalahari desert, Victoria desert, Gibson desert.

Medditerean type of climate:

- Aso called warm temperate western margin.
- In summers ITCZ shifts northwards bringing offshore trade winds with no precipitation.
- In winters ITCZ shifts southwards bringing onshore westerlies and causing good rainfall.
- In winter they are also under the influence of temperate cyclones causing rainfall.
- It is known for its characteristic dry summers and wet winters.
- Summers are not very hot.
- Vegetation is drought-resistant, deep-rooted short bushes.

- Examples-Grapes, are olives, oranges, and other citrus fruits.
- Regions- California, Central Chile, Southwestern part of South Africa, Southwest, and Southern Australia.

Steppe type of climate:(7:34:40 PM):

- Warm temperate continental type.
- Precipitation is higher than in tropical grasslands due to the influence of temperate cyclones during winters.
- During summers the precipitation is through convection resulting in moderate rainfall throughout the year.
- The grass is short and green.
- Soil is very fertile.
- They are known for large-scale commercial agriculture and animal rearing.
- Regions-Steppes of Central Asia, Prairies in North America, Pampas in South America, Velds in south Africa, Downs in Australia, etc.

China Type of Climate:

- warm temperate eastern margin.
- Precipitation is throughout the year.
- In summer, they are affected by tropical cyclones and in winter they are under the influence of temperate cyclones.
- Summers>> Not too hot, Winters>> Not too cold.
- Vegetation is dense and luxurious.
- Deciduous and evergreen species.
- Regoins>> Southeastern USA, SE Brazil, South Eastern China, SE South Africa, SE Australia.

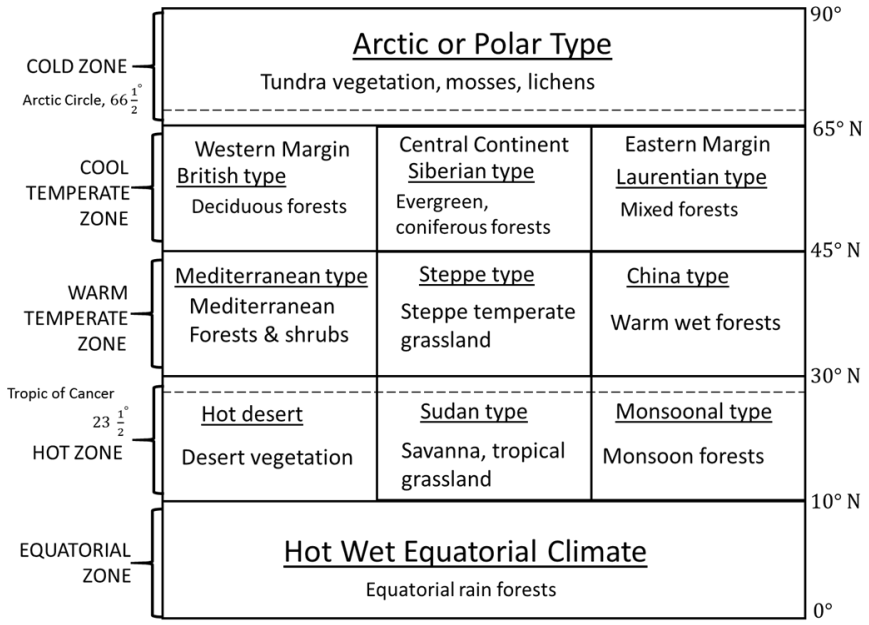
Laurentian type of climate:

- Cool temperate eastern margin.
- Temperature ranges between 20 degrees in summers to below zero degrees in winters.
- Rainfall> throughout the year but more in summers.
- Under the influence of temperate cyclones.
- Vegetation- Upto 50 degrees N and S is deciduous and broad-leaved and beyond that it is coniferous.
- Regions-Eastern Canada, NE USA, Northern China, Korea, and Japan, SE Argentina.

Next class-Continuation of the climatic regions.

Geography Class 30

Brief Discussion of Previous Class (5:09 PM)



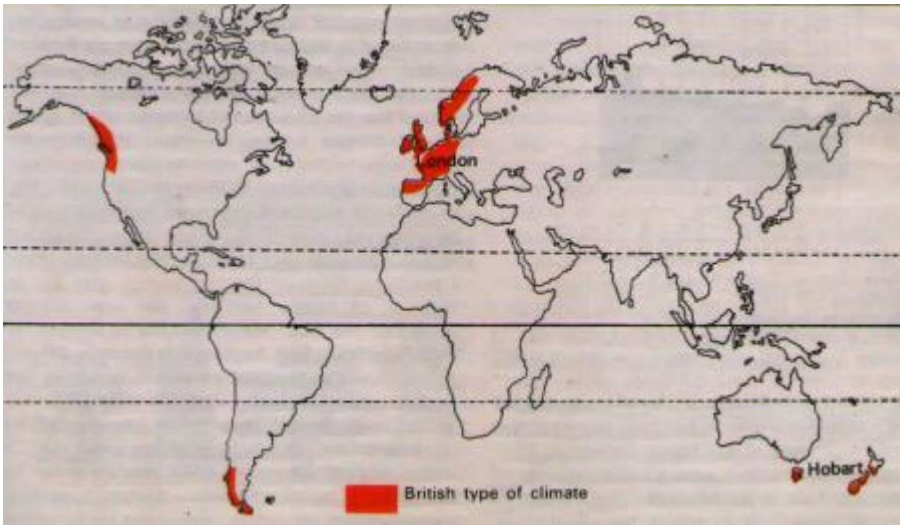
Siberian Type (5:15 PM)

- This is also called cool temperate continental

-
- Summers are hotter, and winters are extremely cold and dry.
- This results in anti-cyclonic circulations.
- The rainfall is through the convection in summer, and through the snowfall in winter
- Vegetation is coniferous and is in the single largest stretch of vegetation on the earth.
- It is low in biodiversity but high in economic value.
- It is the only climate that is exclusively present in the northern hemisphere.
- Regions: Siberia, Russia, Poland, Scandinavia, Canada, Alaska

British Climate(5:30 PM)

- Cool Temperate Western Margin

- 

- They are under the influence of Offshore westerlies, throughout the year.
- This causes precipitation which is moderate throughout the year.
- They are under the influence of warm ocean currents, which keep their ports ice-free even during winter.
- It is known for its characteristic mild winter and cool summer.
- Vegetation is a mixed type, i.e. mixture of broad-leaf and coniferous.
- Regions: United Kingdom, Norway, the western coast of Western Europe, Western Canada, Southern Chile, New Zealand, and Tasmania.

Polar type of Climate (5:40 PM)

- It is also known as the Tundra type of Climate.
- It is covered with ice throughout the year.
- During summers when the snow melts, vegetation grows stunted, like mosses, sedges, and lichens.
- The rest of the region is covered with permafrost.
- Regions: Any region beyond the Arctic and Antarctic circles.

Koeppen Classification

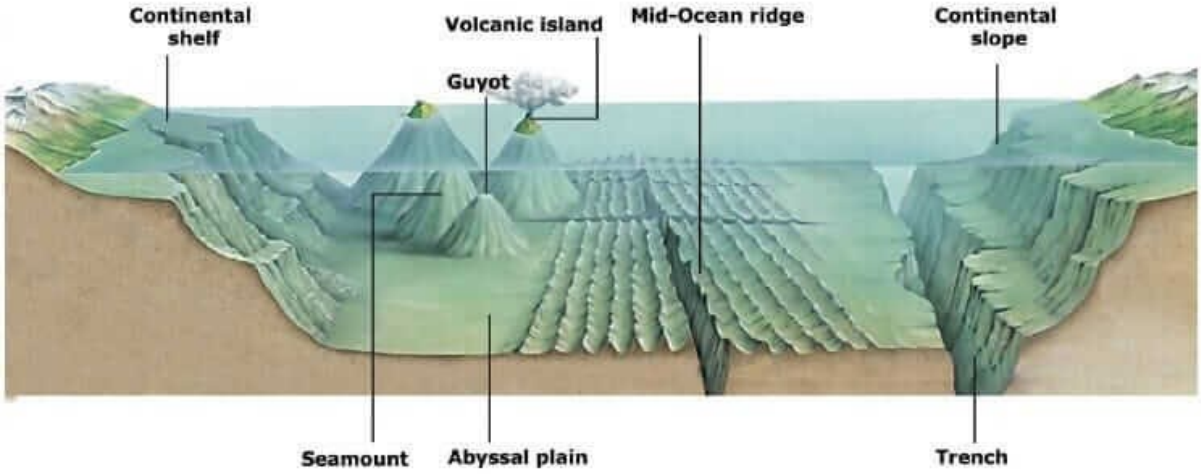
Table 12.2 : Climatic Types According to Koeppen

Group	*Type*	*Letter Code*	*Characteristics*
A-Tropical Humid Climate	Tropical wet	Af	No dry season
	Tropical monsoon	Am	Monsoonal, short dry season
	Tropical wet and dry	Aw	Winter dry season
B-Dry Climate	Subtropical steppe	BSh	Low-latitude semi arid or dry
	Subtropical desert	BWh	Low-latitude arid or dry
	Mid-latitude steppe	BSk	Mid-latitude semi arid or dry
	Mid-latitude desert	BWk	Mid-latitude arid or dry
C-Warm temperate (Mid-latitude) Climates	Humid subtropical	Cfa	No dry season, warm summer
	Mediterranean	Cs	Dry hot summer
	Marine west coast	Cfb	No dry season, warm and cool summer
D-Cold Snow-forest Climates	Humid continental	Df	No dry season, severe winter
	Subarctic	Dw	Winter dry and very severe
E-Cold Climates	Tundra	ET	No true summer
	Polar ice cap	EF	Perennial ice
H-Highland	Highland	H	Highland with snow cover

Oceanography (6:03 PM)

- Ocean bottom Topography
- Ocean temperature
- Salinity
- Ocean Deposits
- Coral reefs
- Oceanic Currents and Circulations
- El Nino and related phenomena
- Tides

Ocean Bottom Topography (6:24 PM)



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- **<u>Continental shelf</u>**
- (a) These are submerged shallow portions of the continental crust with depths between 150-200 m
- (b) It is rich in sunlight, oxygen, and nutrients, which favours the growth of plankton.